

Washing Machine Repair

Chapter 2

PROBLEMS COMMON TO ALL WASHER BRANDS

Washing machine designs vary widely, but there are *some* things that *all* washers have in common. For example, all washers have an electric motor. All washers have both spin and agitate cycles. And since both cycles are driven by the same electric motor, all washers have some sort of mechanism to change between the two.

All washers must also have a way of filling the tub with wash water and a way of draining out used wash water.

And incidental to this, all washers must have a way of controlling water level in the tub, to prevent spillage by overfill or by centrifugal force during the spin cycle.

All washers must (by law) have a mechanism that brakes the spinning basket at the end of the spin cycle, or else a lid lock to prevent the lid from being opened during or shortly after the spin cycle.

And last but not least, all washers must have a timer that controls and coordinates the start, stop and duration of the various cycles.

2-1 CYCLES

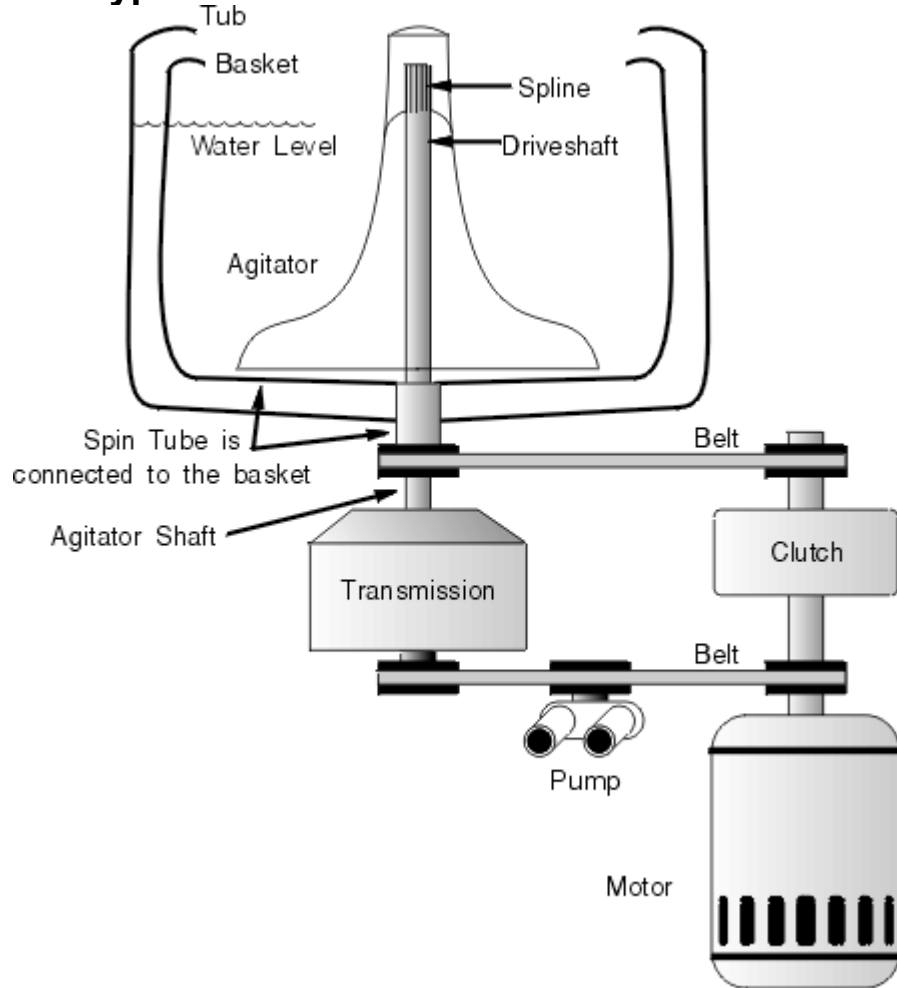
FILL CYCLE

During the FILL cycle, a solenoid-operated water mixing valve opens and allows hot or cold water (or both) to enter the tub. There is no pump operating at this time; the tub fills strictly from house pressure.

Similarly, there is no heater in your washer; the heat comes from the water heater in your home.

When the water in the tub reaches a certain level, a pressure switch closes the water solenoid valve(s).

Figure G-1: Typical Drive Train



WASH/RINSE

(AGITATION) CYCLE

After the water valve closes, an electric motor starts which drives the transmission, sometimes through a belt, and in some cases through a clutch arrangement, too. The transmission converts the rotary motion of the motor to the back-and-forth motion of the agitator. A driveshaft extends from the top of the transmission to the agitator, where it is connected to the agitator, usually by a spline. (See Figure G-1.)

Agitation will continue for a certain amount of time, which is controlled by the timer. During agitation, some washers use their pump to circulate water, sucking it from the bottom of the tub and pumping it to the top of the tub. The pump is driven by the same electric motor.

Also during the agitation cycle, bleach or softener may be automatically added. This is usually done by a solenoid valve that allows some of the recirculated water to flush out the bleach or softener dispenser. In some models there is no water circulation involved; the solenoid simply opens a valve or door that lets the dispenser contents drop into the wash water. The timer tells this solenoid valve when to open.

SPIN AND DRAIN CYCLES

After agitation comes a drain cycle, sometimes combined with a spin cycle. During the drain cycle, the pump sucks water from the tub and sends it down the drain. During the spin cycle, the same motor that drove the agitator now drives a spin tube

which is concentric with the agitator shaft. (See Figure G-1) The spin tube spins the basket, slinging excess water out of the clothes by

centrifugal force. There is a clutch arrangement which allows the basket to come up to speed slowly. This prevents a heavy load from being thrown onto the motor suddenly. It also allows a "pre-pump" action; the water has a chance to drain out of the tub before the basket gets up to speed, preventing the water from overflowing the edge of the tub by centrifugal force.

Some brands have a partial drain cycle only, then refill and agitate again. Some start spinning and draining at the same time. Some only drain until the water reaches a certain level, then start spinning. Most brands have lid switches that prevent the basket from spinning when the lid is open. Some brands have a lid lock that prevents you from opening the lid when the basket is spinning.

At the end of the spin cycle, or whenever the lid is lifted, most models have a braking arrangement that stops the tub from spinning. This helps to prevent people from accidentally sticking their hands into a spinning basket.

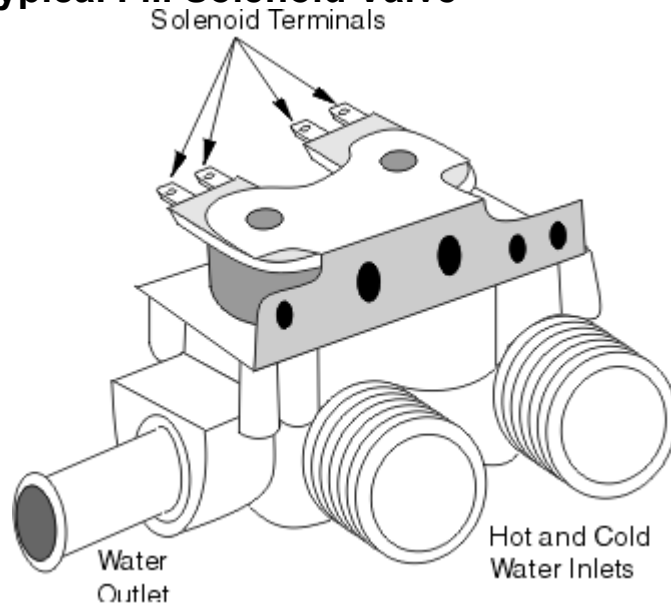
CAUTION: NEVER BYPASS THE LID SWITCH, EXCEPT FOR TEST PURPOSES. IF IT IS DEFECTIVE, INSTALL A NEW ONE. THEY ARE THERE FOR A REASON. NO JOKE: I PERSONALLY HAVE A NEPHEW WHOSE ARM WAS TORN OFF (AT 2 YEARS OLD) BECAUSE OF A BYPASSED LID SWITCH!

2-2 FILL SYSTEM

The basic components of the fill system are the hoses, the fill valve, and the pressure or float switch.

The fill valve (Figure G-2) is simply a solenoid valve that opens when activated and allows hot and/or cold water to flow into the tub. Most modern washers use dual solenoid valves, which have both hot and cold solenoids in one valve body. When warm water is desired, both valves open to mix hot and cold.

Figure G-2: Typical Fill Solenoid Valve

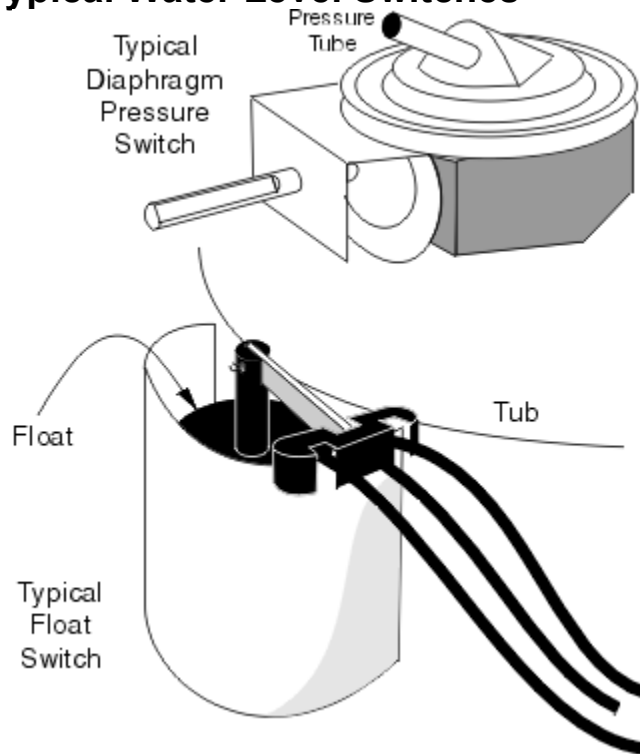


In the newest digital washers, water temperature may be controlled by a "thermistor." A thermistor is a variable resistor; a resistor whose resistance varies according to what temperature it senses. The thermistor's signal goes to the washer's control board (computer,) which opens or closes hot and cold water valves to control the incoming water temperature.

When the water in the tub reaches the desired level, the pressure or float switch closes the circuit to the fill valve. Float switches are pretty rare; you'll find them only in older washers. A diaphragm-type pressure switch is more common. Typical float and pressure switches are shown in Figure

G-3. A few old washers used a pressure switch mounted on the BOTTOM of the tub; these are known as water weight switches. They are rare.

3Figure G-3: Typical Water Level Switches



DIAGNOSIS

If your washer is not filling properly, or is overflowing, there could be several reasons.

SLOW OR NO FILL, OR WATER TOO HOT

OR TOO COLD

If your washer is filling very slowly or not at all, or the water temperature is always too hot or too cold, check the fill hose and valve strainers.

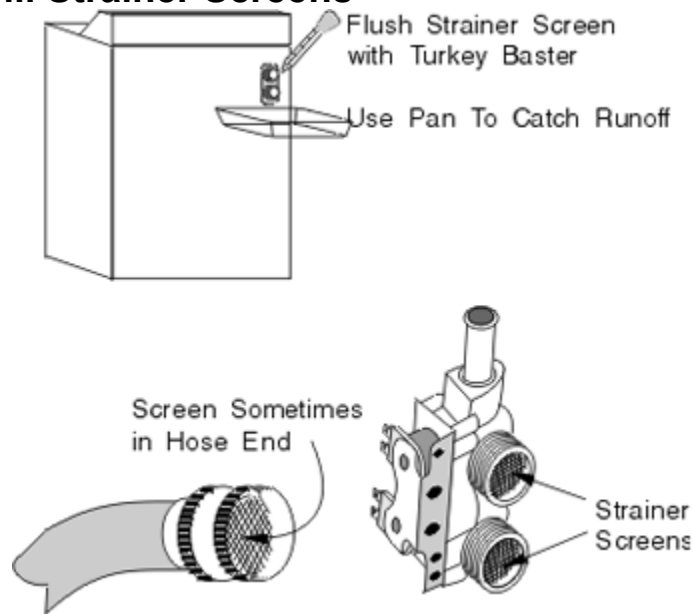
These are little screens placed in the fill hose and/or water valve to prevent rust and scale from your house's piping system from getting into

the water valve. The strainers can get clogged up over time and prevent water flow.

Shut off the water valves and remove the hoses. Look into the both ends of each hose and into the water valve mounted on the washer. In at least one of the three places you should see a strainer screen. (Figure G-4) If it is clogged, you can try cleaning it out with a toothbrush and/or a turkey baster. If you cannot clean the screen sufficiently, you may need to replace it.

In some instances, the screen is non-removable, and you will need to replace the hose or valve. Neither is very expensive.

Figure G-4: Fill Strainer Screens



When re-installing the hoses, always use new hose washers. Also, take care not to overtighten the hose on the plastic threads of the solenoid valve; tighten just enough to stop it from leaking.